

Sheet G09 - Differential Forms

Solutions for the theoretical and practical part via eCampus by Mo, 15.12.2025, **10:00**.

Practical Assignment

No practical assignment this week.

Theoretical Assignment

Assignment 1) Directional Derivatives

(3Pts)

Suppose that $a, b \in \mathbb{R}$, p a point on manifold $\mathbb{R}^3$, $v_p, w_p \in T_p(\mathbb{R}^3)$ and $f, g : \mathbb{R}^3 \rightarrow \mathbb{R}$

- Show that $(av_p + bw_p)[f] = av_p[f] + bw_p[f]$
- Show that $v_p[af + bg] = av_p[f] + bv_p[g]$
- Show that $v_p[fg] = v_p[f]g(p) + f(p)v_p[g]$ (Leibnitz rule)

Assignment 2) Differential One-Forms

(4Pts)

Consider a vector

$$v = \begin{bmatrix} -1 \\ 3 \\ -4 \end{bmatrix} = -\frac{\partial}{\partial x^1} + 3\frac{\partial}{\partial x^2} - 4\frac{\partial}{\partial x^3}.$$

- Show how the differential forms dx^1, dx^2, dx^3 act on it, by computing $dx^1(v), dx^2(v), dx^3(v)$
- Compute $(2dx^1 - 3dx^2 + 5dx^3)(v)$
- The following real-valued functions on the manifold $\mathbb{R}^3$: $f(x, y, z) = x^2y, g(x, y, z) = \frac{x}{2} + yz, h(x, y, z) = x + y + 3$ define at each point $p = (x, y, z) \in \mathbb{R}^3$ the one-form ϕ by $\phi_{(x,y,z)} = f(x, y, z)dx + g(x, y, z)dy + h(x, y, z)dz$. Note that ϕ defines a different 1-form at each $p \in \mathbb{R}^3$
 - Write down the one-form ϕ_p at the point $p = (1, 2, 3)$
 - Apply the one-form ϕ_p to the vector at the point $p = (1, 2, 3)$

$$v_p = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}_p = \left(2\frac{\partial}{\partial x} - \frac{\partial}{\partial y} - 2\frac{\partial}{\partial z} \right)$$

by computing $\phi_p(v_p)$

Assignment 3) Wedge Product and Volumes

(4Pts)

The projection of a vector v onto the $\partial_{x_1}, \partial_{x_2}$ -plane is given by $dx^1(v_p) = v_i$ and $dx^2(v_p) = v_2$. The area of the parallelepiped spanned by these projections is given by $dx_1 \wedge dx_2(v, w)$. As an example, suppose that

$$v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ and } w = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

- Compute $dx_1 \wedge dx_2(v, w)$, $dx_2 \wedge dx_1(v, w)$, $dx_1 \wedge dx_1(v, w)$, and $dx_2 \wedge dx_2(v, w)$.
- Draw a picture of what is happening.
- Compute the Wedge product between $\alpha = 3dx - 7dy$ and $\beta = 2dx + dz$.
- List all elements of the basis of $\bigwedge_p^3(\mathbb{R}^4)$

Assignment 4) Exterior Derivative

(4Pts)

- Let $\alpha \in \bigwedge_p^1(\mathbb{R}^2)$ with coefficients $\alpha = (a_1, a_2)$, where $a_i : \mathbb{R}^2 \rightarrow \mathbb{R}$. Compute $d\alpha$. Write down all intermediate steps.
- Let $f \in \bigwedge_p^0(\mathbb{R}^n)$ and twice continuously differentiable. Verify that $ddf = 0$.

Assignment 5) Hodge Star

(3Pts)

In the lecture, we have defined the Hodge star operator

$$\star : \bigwedge_p^k(\mathbb{R}^n) \rightarrow \bigwedge_p^{n-k}(\mathbb{R}^n)$$

that maps a k -form α to a $n - k$ -form $\star\alpha$ which is implicitly defined by

$$\alpha \wedge \beta = \langle \star\alpha, \beta \rangle_{\wedge} \cdot dx^1 \wedge \dots \wedge dx^n. \quad (1)$$

where $\langle \alpha_1 \wedge \dots \wedge \alpha_k, \beta_1 \wedge \dots \wedge \beta_k \rangle_{\wedge} = \det(\langle \alpha_i, \beta_j \rangle)$ denotes the inner product on $\bigwedge_p^k$ that was defined in the lecture. For this exercise, assume that the metric tensor G is the identity.

- For $n = 3$, show that

$$\begin{aligned} \star(dx) &= dy \wedge dz \\ \star(dy) &= -dx \wedge dz \\ \star(dz) &= dx \wedge dy \end{aligned}$$

using the definition above.

Hint 1: Use the general form $\beta = b_1 dx \wedge dy + b_2 dy \wedge dz + b_3 dz \wedge dx$, to show that Equation 1 is fulfilled.

Hint 2: Let $\sigma_i \in \bigwedge_p^k(\mathbb{R}^n)$ denote the standard basis of k -forms and the k -forms α and β are given as $\alpha = \sum_i a_i \sigma_i$ and $\beta = \sum_i b_i \sigma_i$, then the dot-product between two k -forms under the identity metric can be computed as $\langle \alpha, \beta \rangle = \sum_i a_i \cdot b_i$.

- Let $\alpha = a_1 dx + a_2 dy + a_3 dz$ be a 1-form in $\mathbb{R}^3$. Calculate $\star\alpha$.

Good luck!